GIDEON GYIMAH

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RESEARCH INTERESTS

Evaluation validity for generative models outside the distribution they were built on — negative controls, metric failure, and the reporting of null results. Large language model and agent systems for public-interest communication in low-resource languages, particularly African languages largely absent from mainstream NLP. Factuality and hallucination in retrieval-augmented generation, measured under real deployment: domain shift, adversarial queries, and failure-mode characterisation. Efficient adaptation (LoRA, PEFT) where labelled data is scarce or culturally underrepresented.

EDUCATION

BSc, Electrical and Electronic Engineering — 2019 – 2023

University of Energy and Natural Resources (UENR), Sunyani, Ghana

CGPA: 2.66 / 4.0 (Second Class, Lower Division) · **Thesis:** Electrical Load Forecasting for the Valco Aluminium Smelter using Machine Learning (github repo)

English is the sole medium of instruction and examination at UENR. Final-year performance is materially stronger than the cumulative figure: final-year GPA 3.05 over 39 credits (Semester 1: 3.30), with Research Project I and Research Project II graded B+ (3.50) and A (4.00). Mathematics coursework: A in Engineering Mathematics II, B+ in Linear Algebra for Engineers, B+ in Numerical Methods for Engineers.

PUBLICATIONS

– **Guidance, Not Adaptation: What Makes an Open Text-to-Music Model Sing isiZulu.** *Accepted* — 1st Africa in AI Workshop, NeurIPS 2026 (Springer LNCS). Accepted October 2026.

First published evaluation of any African language on an open text-to-music model. Adapted ACE-Step 1.5, a 2B-parameter flow-matching diffusion transformer, to isiZulu with a LoRA trained on 10.95 hours of Maskandi, Kwaito, Amapiano, Isicathamiya, Mbaqanga and Gqom; 3,214 generations over 60 prompts with seeds matched across every condition. **Finding 1:** classifier-free guidance, not the language adaptation, controls whether the model sings isiZulu — composite lyric error falls from 0.957 to 0.459 and coverage rises from 0.072 to 0.680 across guidance 1–15 (guidance 1 vs 7: $d_z = -2.57$, permutation $p < 10^{-5}$, $n = 60$). **Finding 2:** the caption-retrieval control built to validate the genre metric was itself broken in three independent ways, each pushing towards a false negative, and none of them raised an error. Corrected, the correct caption ranks 6.6 places above a mismatched one (95% CI [3.0, 10.1], $p = 0.0008$, clustered over 63 captions), where the binarised statistic used first found nothing. Corpus schema, prompt set, pre-registration and evaluation harness released.

RESEARCH EXPERIENCE

AI Research Scientist — African Music AI — Nov 2025 – Mar 2026

*Wits AI & African Music Project, MIND Institute, Wits University, South Africa · In collaboration with artist **Umlilo** (Future Kwaai Records, South Africa) · official announcement · project profile*

– **Collaboration:** Worked as the AI engineer alongside artist Umlilo (multidisciplinary creative director, Future Kwaai Records) to co-develop *Zazi* — a real-time voice, rhythm, and storytelling interaction system rooted in Zulu musical identity.

– **Problem:** How can generative models capture the tonal and culturally specific features of Zulu music when only a small audio corpus is available and standard pre-training data is almost entirely Western?

- **Approach:** Built and adapted a generative music model (AceStep architecture) using LoRA fine-tuning on a curated Zulu audio dataset; applied targeted data augmentation to partially compensate for corpus size limitations.
- **Supervision:** Research conducted under the supervision of Prof. Christo Doherty (Acting Angela & David Fine Chair of Innovation, Wits Innovation Centre) and the MIND Institute research team.
- **Outcomes:** Model weights and live training logs published openly on HuggingFace — findings contributed to the Wits MIND Institute's African AI research programme. Full training transparency maintained via public TensorBoard logs (TensorBoard · model hub).

Lead AI Researcher — Multilingual NLP for Low-Resource Languages

2025 – Present

UNESCO Ghana, Japan-Funded Disaster Resilience Project (Remote)

- **Problem:** Can end-to-end NLP pipelines for speech recognition, language understanding, and synthesis be built for Ghanaian languages — Twi, Ewe, Hausa, and Ga — where labelled data is essentially absent and no commercial ASR or TTS tools exist?
- **Approach:** Designed and built a multilingual NLP pipeline combining low-resource ASR, intent classification, and TTS; constrained inference to operate on 2G networks without requiring literacy from end users.
- **Outcomes:** Live WhatsApp disaster-resilience chatbot deployed and serving communities across Ghana (wa.me/233204833556 · +233 20 483 3556). One of very few deployed NLP systems operating natively in Ga or Ewe.

AI Engineer — LLM Evaluation & Production NLP Systems

Jan 2025 – Present

Inlaks LTD, Accra, Ghana — Enterprise Fintech & Banking Technology

- Designed evaluation frameworks measuring hallucination rates, groundedness, and failure modes in LLM-powered banking agents — motivated by the challenge of deploying language models in high-stakes, low-error-tolerance domains at scale (10,000+ daily interactions).
- Built multi-agent retrieval-augmented systems (LangGraph); studied how model behaviour degrades under domain shift, adversarial queries, and out-of-distribution inputs — empirical findings directly relevant to LLM robustness research.
- Developed behavioural scoring models for customer churn prediction (92% accuracy); designed voice AI infrastructure handling 1,000+ concurrent sessions with real-time transcription and model routing.

AI/ML Researcher & Teaching Assistant

Aug 2023 – Oct 2024

ACE – Regional Centre for Energy & Environmental Sustainability, UENR + Azubi Africa, Ghana

- **Problem:** Extending automatic speech recognition to Hausa — spoken by 70M+ people across West Africa but with almost no labelled speech data for model training.
- **Approach:** Fine-tuned OpenAI Whisper on a Hausa speech corpus using data augmentation (speed perturbation, noise injection) to partially offset scarcity; evaluated against held-out speakers and noise conditions. Also applied k-means clustering to energy consumption data as part of the centre's sustainability research.
- **Outcomes:** Model and full methodology published openly (Kaggle notebook · HuggingFace models). Mentored 50+ learners across three cohorts in Python, ML, and data analysis.

RESEARCH OUTPUTS & OPEN-SOURCE CONTRIBUTIONS

- **isiZulu Generative Audio Model** (ACE-Step 1.5, LoRA). Wits MIND Institute, 2026. Generative model trained on a low-resource isiZulu audio corpus using LoRA. Weights, training dynamics, and TensorBoard metrics publicly available. huggingface.co/Gyimah3 · zaziai.com
- **Hausa Automatic Speech Recognition** (Whisper fine-tune). ACE/UENR, 2024. Low-resource ASR model with full methodology documented. kaggle.com/gideongyimah/finetuning-hausa · huggingface.co/Gyimah3/models
- **DocuMind — agentic retrieval pipeline for compliance documents**. 2025. Intent detection, query rephrasing, Redis KNN retrieval, FlashRank cross-encoder re-ranking, and generation with source citations.

PII is masked before any chunk reaches the model; multi-tenant with JWT-scoped isolation; images pass through a vision model into searchable text. Abstains — answers "I don't know" rather than fabricating — which is the behaviour the evaluation work above is built to measure. github.com/Gyimah3/DOCUMIND-SS-AGENTIC-RAG

- **SIGMA — human-in-the-loop co-agent**. 2025. The model proposes structured diffs to a business model canvas and a person accepts or rejects each one. Propose, apply, undo, redo, version history, idempotency by proposal hashing, and a full audit trail. An autonomy toggle lets it act without approval, so the operator decides how much autonomy the agent is given — a concrete study in where the human belongs in an agent loop. github.com/Gyimah3/Sigma-prototype-autonomous-agent
- **Agentic RAG System — Domain-Specific Question Answering**. 2024. Earlier multi-tool RAG agent combining internal retrieval with live web search; 1,000+ monthly queries in production. github.com/Gyimah3/UENR-CHATBOT-2 · live
- **CO_2 Emissions Prediction from Satellite Imagery**. AI Ghana Competition, 2024. ML model on Sentinel-5P data for African carbon emissions. 1st Runner-Up, 150+ participants. Zindi leaderboard
- **Electrical Load Forecasting** (BSc Thesis). UENR, 2023. ML models for short-term demand forecasting at the Valco Aluminium Smelter, Ghana. github repo

TECHNICAL SKILLS

Languages	Python (primary); reproducible experiment pipelines; scripting
ML / NLP	HuggingFace Transformers, PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph
LLM Methods	LoRA / PEFT fine-tuning, RAG, hallucination & factuality evaluation, prompt engineering, multi-agent orchestration
Speech & NLP	ASR (Whisper), TTS, multilingual NLP, low-resource adaptation, data augmentation
Infrastructure	AWS (Certified), Docker, FastAPI, PostgreSQL, vector databases (Pinecone, Weaviate, Chroma)
African Languages	Twi (native speaker), Hausa, Ewe, Ga, Zulu, Swahili — research and deployment experience across all six

AWARDS & RECOGNITION

Selected Researcher — Wits AI & African Music Project 2025 – 2026
Competitive selection by the Wits Innovation Centre and MIND Institute for a funded African music AI research programme. (announcement)

1st Runner-Up — AI Ghana CO_2 Prediction Challenge (Top 2 of 150+) 2024
ML model on Sentinel-5P satellite imagery for African carbon emissions prediction. (leaderboard)

1st Runner-Up — ElevenLabs World Conversational Agent Hackathon (Global, Exa Search Category) 2024
Built production-quality voice agent with live search integration. (demo · confirmation)

1st Runner-Up — IndabaX Ghana 2024, Retail Analytics Hackathon 2024
SQL agent processing 100,000+ records across African retail markets. (recognition)

CERTIFICATIONS

AWS Certified Cloud Practitioner (verify) · Data Analyst Professional (verify) · Certified Business Analyst (verify) · Deep Learning Indaba Participation (certificate)

REFERENCES

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